# ELECTROMAGNETIC INDUCTION IN DAILY LIFE

PYQ • 2021 - 2026

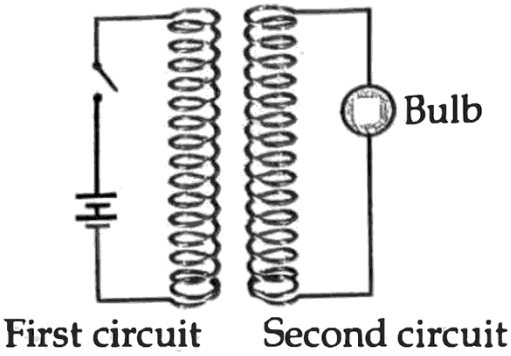
**1.** 2026 **4 MARKS**

(A) Two solenoids are placed side by side. Observe the figure.

(a) The bulb in the second circuit lights up only at the moments when the switch in the first circuit is turned ON or OFF. Why does the bulb fail to glow continuously when the switch remains ON ?

(b) What change should be made to the first circuit, for the bulb in the second circuit to glow continuously ? What is the phenomenon, by which current is produced in the second circuit in this manner ? Explain the phenomenon.

(c) The first coil in this device has 100 turns and the second coil has 200 turns. If 400 V is to be obtained in the second coil, what voltage should be applied to the first coil ?



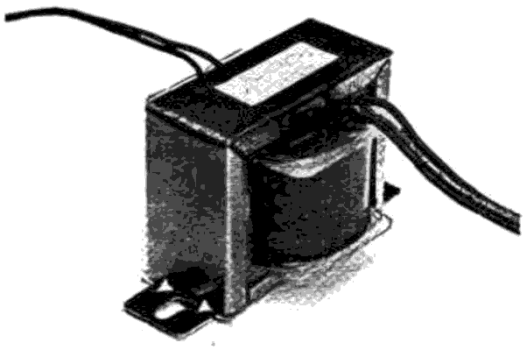

**2.** 2026 **4 MARKS**

(B) (Choice for the question above) Observe the figure.

(a) What is the name of this device. What is its use ?

(b) When 20 V is supplied to the primary coil of this power lossless device, 100 V is obtained in the secondary coil. If a current of 1 A flows through the secondary coil, calculate the current flowing through the primary coil.

(c) What are the differences in the number of turns and the thickness of the wire between the primary and secondary coils of the device, for which you just calculated the current ?

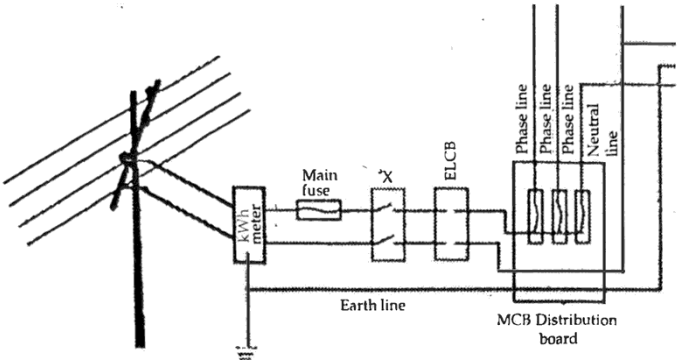


**3.** 2026 **2 MARKS**

A portion of electric circuit diagram of a household wiring is given.

(a) What does the part marked as "X" represent in the circuit ?

(b) Draw a circuit diagram of a branch circuit which includes one three pin socket with switch.



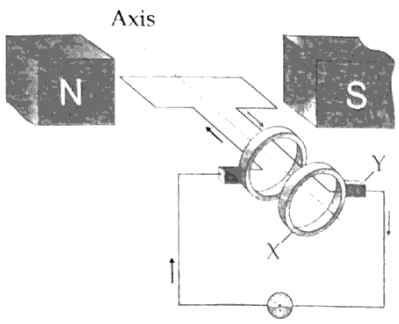

**4.** 2025 **4 MARKS**

The schematic diagram of a generator is given below.

(a) Name the parts labelled as 'X' and 'Y'.

(b) Which type of current is produced as output here ? Draw the graph of output voltage.

(c) State the law that helps to find the direction of current in the armature coil.

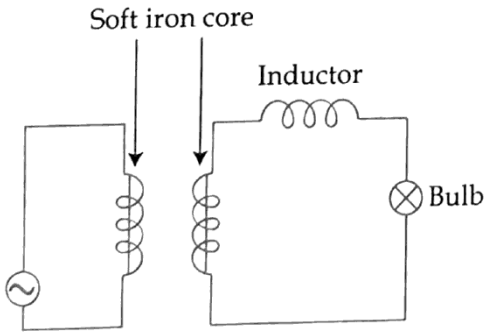


**5.** 2025 **3 MARKS**

Observe the circuit below with two coils placed close to each other. When AC flows through it, the bulb glows.

(a) Why does the bulb glow ?

(b) When a soft iron core is placed inside the inductor, what change will be observed in the intensity of brightness of the bulb. Justify your answer.




**6.** 2025 **2 MARKS**

What is transmission loss in an electrical power distribution system ? What measures have been taken to minimise it ?

**7.** 2024 **4 MARKS**

A transformer has 100 turns in the primary and 1000 turns in the secondary.

(a) Which coil of this transformer is made using thick wire ? Give reason.

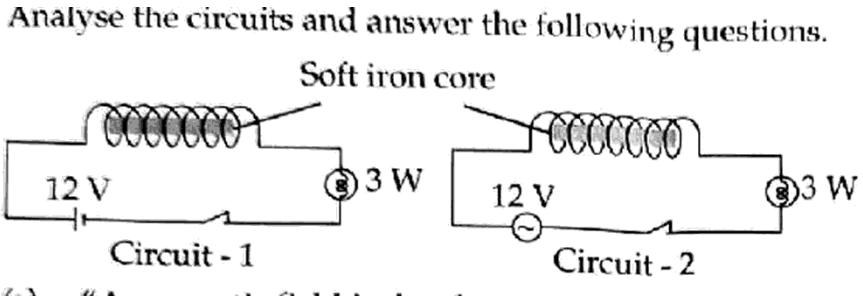
(b) Explain how electrical energy is transferred from the primary to the secondary of the transformer.

**8.** 2024 **3 MARKS**

Analyse the circuits and answer the following questions.

(a) "A magnetic field is developed around the solenoid only in the second circuit." Do you agree with this statement ? Explain.

(b) In which circuit, a continuous emf is induced ? Name and explain this phenomenon.



**9.** 2023 **4 MARKS**

AC generators are used in power stations in our Country.

(a) What is the voltage produced by the generators in our power stations ?

(b) What do you mean by transmission loss ?

(c) Explain how it is minimised.

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**10.** 2023 **3 MARKS**

The current in the secondary coil of a transformer with no power loss is 5 A and that in the primary is 0.5 A.

(a) What type of a transformer is this ?

(b) If the input voltage of this transformer is 240 V calculate the output voltage.

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**11.** 2023 **1 MARK**

When there is a change in magnetic flux linked with a conductor, an emf is induced in it. Name the phenomenon.

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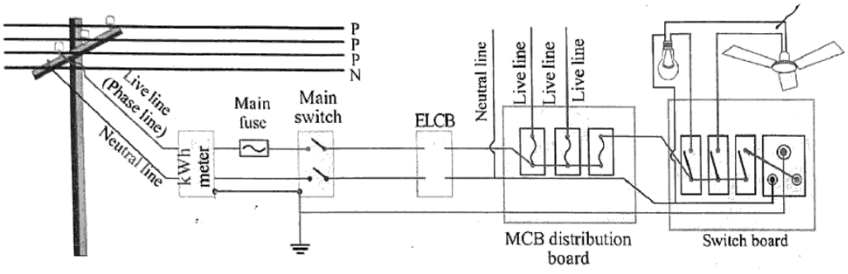
**12.** 2022 **4 MARKS**

Observe the circuit of household electrification.

(a) Which device is used to measure the electrical energy consumed in household circuit ?

(b) Write any two advantages of connecting the devices in parallel in household circuit.

(c) Write the function of ELCB.




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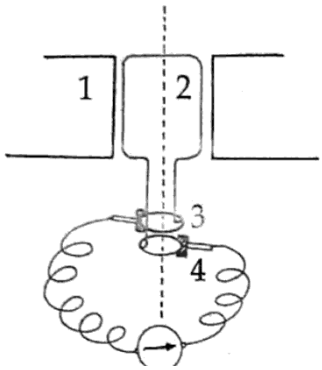
**13.** 2022 **4 MARKS**

Schematic diagram of a generator is given.

(a) Which type of generator is this ? (AC/DC)

(b) Name the parts of this generator marked as 1, 2, 3, 4.

(c) State the working principle of this device.




**14.** 2022 **3 MARKS**

Some characteristics of step up transformers and step down transformers are given below. Select the statements suitable for step up transformers.

(a) Primary voltage is greater than secondary voltage.

(b) Secondary voltage is greater than primary voltage.

(c) Current in the primary coil is greater than that in the secondary coil.

(d) Current in the secondary coil is greater than that in the primary coil.

(e) Thick wires are used in the primary.

(f) Thick wires are used in the secondary.

**15.** 2022 **2 MARKS**

Write any two precautions to be taken to avoid electric shock.

**16.** 2022 **1 MARK**

Secondary coil of a transformer has double turns than that of its primary coil. If the voltage applied in the primary coil is 25 V, what will be the voltage in secondary ?

(25 V, 50 V, 2 V, 12.5 V)

**17.** 2022 **1 MARK**

Which arrangement converts the AC induced in the armature of DC generator into DC ?

**18.** 2021 **4 MARKS**

(a) Write the function of a safety fuse in an electric circuit.

(b) Write any two characteristics of fuse wire.

(c) Write any two precautions to be taken while including fuse wire in a circuit.

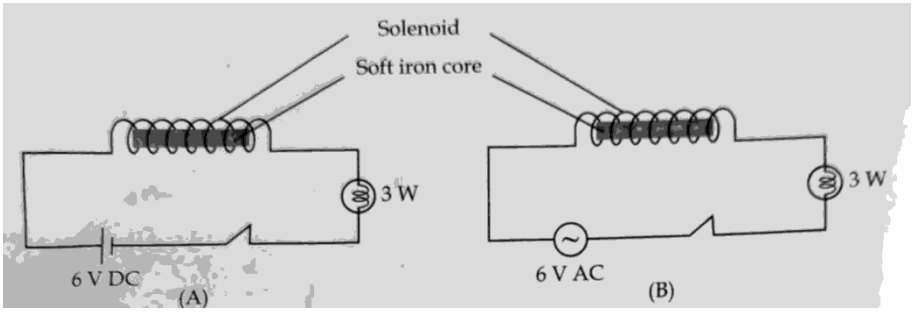
**19.** 2021 **4 MARKS**

Analyse the figure to answer the questions.

(a) In which circuit is a varying magnetic field developed around the solenoid ?

(b) Which bulb in the circuit glows when the switch is kept on ?

(c) Explain whether there is a difference in the intensity of the light of the bulb.




**20.** 2021 **4 MARKS**

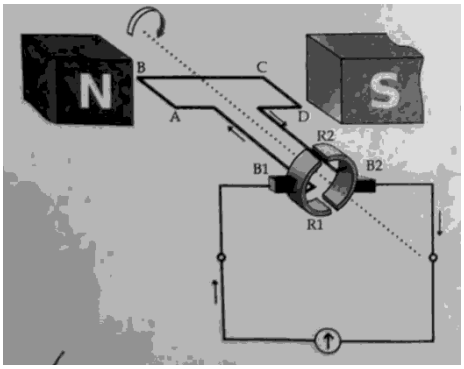
Observe the figure.

(a) Identify the Generator.

(b) Write the working principle of this generator.

(c) Which type of emf is induced in the armature ?

(d) Write the function of split ring commutator used in this generator.

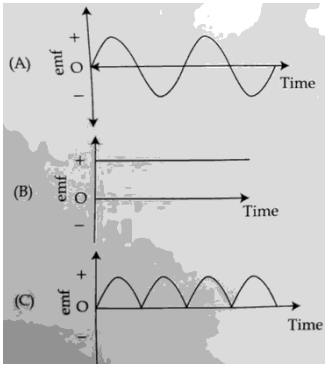


**21.** 2021 **3 MARKS**

The graphical representation of emf obtained from three electrical sources are given below.

(a) Identify any two sources.

(b) Write any two peculiarities of each of the emf shown in the graph (A) and (B).




**22.** 2021 **3 MARKS**

The voltage of electric power generated is increased from 11 kV to 220 kV in power transmission. But the household supply is provided at 230 V.

(a) Write the type of transformer used at the first phase of power transmission.

(b) Write the type of transformer used at the distribution line.

(c) Write any two structural differences between step-up and step-down transformer.

**23.** 2021 **2 MARKS**

Explain the following.

(a) Electromagnetic Induction

(b) Induced emf

**24.** 2021 **2 MARKS**

Write any two relevant first aids to be given in the case of electric shock.

**25.** 2021 **2 MARKS**

Explain the difference between short circuit and overloading.